Data Driven Control Integration Week

Group 8 (MPC)

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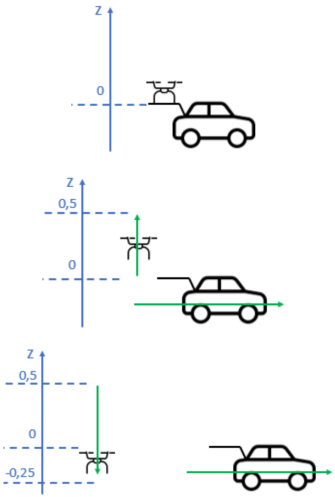
Project Objectives

Available Resources:

- Dataset of vertical acceleration.
- MATLAB/Simulink drone simulator.

Objectives:

- Validate the dataset.
- Identify the model.
- Develop a control law.



Dataset Validation

Cleaning the data

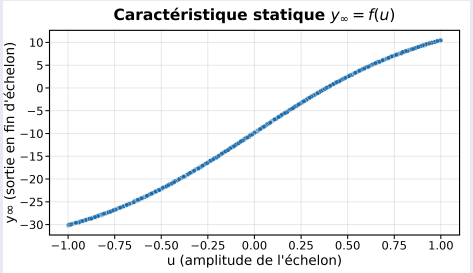
Remove lines that includes:

- Frozen sensor for 2+ seconds
- 5+ diverging values

Normalisation

- Directly in the training

Static characteristic



$y_\infty = f(u)$ after eliminating corrupted runs.

System Modeling

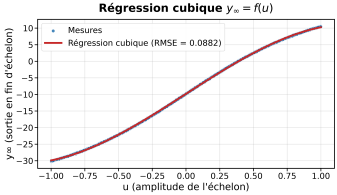
Linear Model: Hammerstein Approach

Static Characteristic $N(u)$

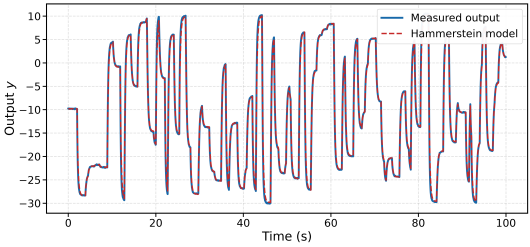
3rd-order polynomial identified on steady-state regimes:

$$v(t) = N(u) = a_3 u^3 + a_2 u^2 + a_1 u + a_0$$

Tau identification



Hammerstein Model Validation



Least-Squares Linear Model Validation (a=0.7836, b=4.9676, c=-2.1222)

System Modeling

Linear Model: Least Square Method

Theorical analysis

Least-squares criterion

$$\hat{\boldsymbol{a}} = \arg\min_{\boldsymbol{a}} \sum_{i=1}^{N} \left(y_{\infty,i} - \hat{y}_{\infty}(u_i)\right)^2$$

